

Grasping the Grammar with Moovy: a Tangible User Interface to Train Linguistic Skills in Children

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ABSTRACT

Linguistic skills are the building blocks of one of the most resourceful abilities of human beings: communication. Compromised linguistic skills significantly reduce the understanding or exchanging of meaningful messages and information. Children with difficulties in this area are often enrolled in extensive speech therapy programs and, nowadays, such issues affect also bilingual children with a migrant background, impacting their quality of life. Our research explores a Tangible User Interface developed on a new paradigm we postulated called "Embodied Argument Movement". In this paper, we present Moovy, an engaging table-top game for rehabilitating challenging morphosyntactic constructs and improving language skills. We performed two pilot studies with 13 children; one to test the User Experience and the other to assess Moovy's efficacy and the paradigm soundness. Preliminary results suggest that Moovy

could make a difference in speech-therapies programs, fostering children to continue their therapy willingly and successfully.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction paradigms**; *Accessibility technologies*; • **Applied computing** → **Interactive learning environments**.

KEYWORDS

tangible interaction, children, smart toys, literacy, RFID, user interfaces, diy, serious games

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1 INTRODUCTION

Language disorders (LD) are impairments that limit the processing of linguistic information and can affect different structural constraints such as grammar (syntax and/or morphology), semantics (meaning), or pragmatics [22]. If not treated, problems in language development might persist through school age (or beyond) and can severely impact the quality of life. Awareness about these types of disorders has increased significantly in recent years and there is multidisciplinary interest among clinicians and researchers to enrich both the comprehension of LD and the rehabilitation tools [11]. According to literature, almost 7% of young children experience a language disorder [5, 8, 12, 31] and among them, Developmental Language Disorders (DLD) affect about 1 out of 15 [7]. DLD subjects have persistent and significant language impairments [26] and, although every child is unique, some deficits are more common in this population: troubles using proper grammar or sound [7], reduced vocabulary [38], and problems with specific morphosyntactic structures such as *clitic pronouns* ('Paola opens the book. Then she *it*^(clitic pronoun) leafs through. '), *passive form* ('The book is leafed through (by Paola).') and *relative propositions* ('The girl that leafs through the book.').

Children with DLD are not the only ones undergoing the aforementioned difficulties. Several studies have shown that also bilingual children with a migrant background, also called sequential bilingual children, experience analogous challenges [6, 40, 41].

Among the deficits cited above, clitic pronouns and passive sentences share one important similarity: an *argument movement*, i.e. a movement of a verb argument towards the left periphery of the sentence (as explained in section 2.2.1). The core idea behind our work was to transpose the grammatical "argument movement" into a physical action that takes place in front of the child's eyes and in which (s)he is actively involved. This idea was then transposed into a new paradigm, that we called *Embodied Argument Movement*, theorized from an extensive analysis of the literature on how the brain processes language and action (more detailed description is provided in section 2.2). The correlation between movements and cognitive processes is widely known and supported also by modern theories of Embodied Cognition [37, 43] and there is a variety of evidence with respect to the role that movement plays in learning [43]. Grounded on these theories were designed a great amount of Tangible User Interfaces (TUI) that allow a direct interaction with digital systems. Through these interfaces, children are encouraged to manipulate physical objects and to receive contextual, immediate and automatic reinforcement feedbacks. Numerous studies corroborated TUIs advantages in strengthening collaboration between children through play [30], facilitating learning and decision-making by taking advantage of the instinctive human ability to grasp objects and materials [20], increasing engagement [25], assessing or training memory skills [4] and fostering problem solving [18].

In this paper we describe the "*Embodied Argument Movement paradigm*" and the design and implementation of a Tangible User Interface based on it. Moovy is an engaging tabletop game that allows children to rehabilitate challenging morphosyntactic constructs and improve linguistic skills while engaged in a playful activity. The paper also reports results of two pilot studies we performed

with 13 children; one to test the User Experience and the other to assess Moovy's efficacy and the paradigm soundness. The findings are preliminary and not conclusive but provide some encouraging evidences about Moovy's potentiality to become an effective tool for speech-therapy programs.

The rest of the paper is organized as follows: Section 2 provides an overview of the theoretical background, Section 3 describes Moovy, Section 4 presents the preliminary empirical studies and discusses the main results. Section 5 draws conclusions.

2 THEORETICAL BACKGROUND

2.1 Language and Action

The relationship between language and action has been the object of an extensive investigation [33]. Understanding this relationship would be an invaluable source of information on how the brain evolved and how it works [29]. To date, empirical research has focused largely on a lexical semantic domain, i.e. the significance of action words and on how they are portrayed in our sensory-motor systems [27, 36], more than on structural and syntactic aspects. However, syntax and the sensory-motor systems tap onto the same Broca's area, suggesting a possible convergence between these domains [13, 14]. From an evolutionary point of view, positive evidence that syntax-based representations are partially shared by motor and linguistic systems, imply that, in the early brain, the evolution of syntax has capitalized and built upon parts of a "pre-existing syntax" used by the motor system [2, 15, 36]. According to Gallister [17], actions reveal an argumentative structure that is comparable to that of verbs in the spoken language, especially in relation to subjects and objects. Knott [23] suggests that the "logical form" of a sentence could be understood as a description of the sensory-motor processes involved in the episode it describes and that many of the syntactic principles of sentences indeed have a sensory-motor origin. At the same time, actions are structured in a way that resembles the hierarchical embedding that characterizes the syntax of language. The structure of a certain motor event can be decomposed into goal oriented sub-events that can be variable binding and have long-distance dependencies [21]. In an article published in 2021, Thibau et al. [39] report the results of their work with functional magnetic resonance imaging (fMRI). They demonstrate that tool use and language share a common neural pattern in the basal ganglia. The paper's findings reveal the existence of a *Supramodal Syntactic Function* that is shared between language and motor pathway processes. Researchers also demonstrate that such shared neurofunctional resources also allow for cross-domain transfer [39]. Basically, training of motor processes improves linguistic processes and vice versa.

2.2 Embodied Argument Movement Paradigm

Building on the aforementioned background, we attempted to develop and posit a more application-oriented paradigm that leverages on the above-mentioned *Supramodal Syntactic Function*. The *Embodied Argument Movement Paradigm* theorizes that, in the context of Language Disorders and acquisition, it is possible to create very effective practice if such training is built on the manipulation of objects through body movements that represent argument operations. A representative example of this are clitic pronouns and

passive sentences, described in detail in the next section. These sentences contain a hierarchical organization of information, called argument structure, that must be "rearranged" to allow the subject to understand the sentence meaning. The syntactic "rearrangement" is analogous to the motor "rearrangement" that the subject should perform. This because, in addition to a common neuronal sub-layer, there is a parallelism between the structure of simple action and that of a sentence[36]. In clitics and passives the movement is syntactic and is a perfect benchmark to verify the robustness of the paradigm itself. The possible benefits deriving from the application of this paradigm could be twofold: on the one hand, by targeting the same brain area, it would allow a stronger and more focused stimulation; on the other, it would offer multiple-channels stimuli of different nature of the same syntactic structures that could foster comprehension and elaboration.

2.2.1 Argument Structure: Clitics and Passive sentences. In linguistics, an argument is a unit of syntactic structure (phrase) that appears within a sentence in relation to the verb [1, 35]. Typical syntactic arguments are the subject and the direct object, and the linear canonical argument order in Italian (and English) is subject - verb - object [9]. However, in these languages there are exceptions to this order, in which an argument can move in a pre-verbal or post-verbal position [16]. An argument movement is the movement of a verb argument in a different position with respect to the canonical one. Object clitic pronouns and passive sentences in Italian are morphosyntactic structures that are largely influenced by what is outlined above (English passives too). They both share structural analogies: in particular, the verb argument moves and occurs in an uncanonical pre-verbal position. Object clitic pronouns are morphemes filling a verb direct object syntactic function that denote referents previously introduced in the discourse: "The wizard sees the witch. Then, he *her^(cliticpronoun) hugs (her)".

Passive sentences are syntactic constructions where the direct object (complement) of a transitive verb works as subject of the sentence: "The witch is hugged (by the wizard)". The use and the comprehension of object clitics and passives is particularly important to understand the informational structure of the discourse: in particular, what is the argument of the discourse and what is the subject of the verb. The role of these structures in mastering a language is then central; new instruments to enhance their acquisition are therefore essential for both DLD and sequential bilingual children.

3 MOOVY

3.1 Moovy at a Glance

Moovy is a TUI designed to mimic a real board game. The components (Figure 1) of the game are: A: a tabletop panel; B: an interactive board; C: 36 tagged cards (with 108 types of stimuli in different forms - toys, pictures or written words); D: an App. The Interactive board is the main touchpoint since it provides active feedback and hosts children's inputs. It can give luminous signals together with auditory feedback. It was designed to be as minimal as possible, so that when idle, children are not distracted by it.

3.2 Playing the Moovy game

The game has some very basic and simple rules. Moovy is a fun but spiteful buddy and will tell sentences that always contain one subject and one object among the cards on the table-top panel. However, the sentences he chooses to say can be tricky to understand. The purpose of the game is not to be fooled and place correctly as many cards as possible on the board. When the children completes one activity, they will beat Moovy and the board will glow with rainbow colors emitting a cheerful song (fig. 2). Before getting started with the game, it is necessary to place the board and the stimuli on the table-top panel and launch the activity through the tablet. Once an activity has been selected, the board will reproduce the sentences and provide a feedback every time a card is placed on it. The game interaction has been designed to be focused only on the table-top panel. The webapp acts as a controlling remote for the therapist, and the user interacts only by means of the TUI, thus echoing a real board game experience.

3.2.1 Activities. The assets proposed through Moovy are comprehension tasks. There are two categories: clitic pronouns and passive sentences. Five (5) different activities of sixteen (16) pairs of sentences each are available within the categories; each of the categories holds 68 pairs (136 sentences) for a total of 272 sentences available. The child hears a sentences pair presenting the targeted morphosyntactic construct. Once they hear the first sentence, they are asked to place the subject and object on the board by arranging the arguments of the sentence in the order in which they heard them. For example they will hear "First the child eats an apple" and their task is to place the child card on the board followed by the apple card. If the placement is correct, the board will light up the box where the card is placed in green and, if it is wrong, it will light up in red. Once the correct card has been placed on the board, the board will tell the second sentence: "Then the apple is eaten by the



Figure 1: Moovy's table-top panel, interactive Board, stimuli and webapp



Figure 2: Left Image: Moovy game; Right image: A researcher and a child playing with Moovy

grandfather". At this point the child should take the apple card they previously placed on the board, put it on the line below, then place the *grandfather* next to it. The box, from which the *apple* item has been taken, will light up in white, marking a "ghost trace" of the change of position, that is the "argumentative movement" (Figure 2, left image).

3.3 Technical Implementation

3.3.1 The Board. The board is manufactured with an FDM 3D printer, using robust and recyclable Polyethylene terephthalate glycol (PETG) white filament. The board is sized 220x160x43mm, matching the majority of entry-level 3D printers capabilities. The board is composed of three pieces tighten together with standard M3 screws. On the top surface, there is a grid of 4 rows and 2 columns squared cells, each one sized 53x40mm. Below each cell (inside the board), there is an RFID reader and a LED strip embracing the cell frame. On the lateral back of the board, the chassis exposes a micro USB connector to recharge the LiPo battery (and eventually to upgrade the firmware) and a switch to turn on the board. The board uses a rechargeable 4000mAh LiPo battery, which guarantees more than 8 hours of battery-life, enough for a full working day. The remaining internal components consist of an Espressif ESP32, a powerful microcontroller with WiFi connectivity, a micro SD to store audio files, and an audio speaker.

3.3.2 The stimuli. Each stimulus card is composed of a Tag, which contains the technology for identification, and a Representation that provides the meaning (stimulus) to the child. Velcro straps allow the user to connect the Representation to the Tag. The Tag is composed of a 3D-printed plate and hosts popular RFID key chains. There is a small label on the bottom of each tagged plate to help caregivers during representation replacements. The representation consists of thin plastic plates which support the concrete object, picture or word. Both plates share the same measure, 53x40mm, and they stick together using Velcro adhesives. This choice allows to save the cost of tag duplication, helping to reduce the usage of raw and electronic material, as well as add new representation without altering the underlying technology.

3.3.3 The Application. The Application is the control panel for the caregiver: it allows to start and configure an activity, control the activity during the execution, manage and review the patience and session data. It is a web app, guaranteeing accessibility and easy deployment on every device. The application communicates to the board using Secure WebSocket, hence ensuring privacy. The application collects data about the interaction with the board and the application itself and sends data to a centralized remote database. The caregiver can see aggregated data regarding only their associated children. The more relevant information displayed is the number of sessions done and programmed, the average number of errors, and the latest session date. On the other side, researchers can download the entire database to perform extensive analysis on the whole population.

4 PILOT STUDIES

In this section we describe two pilot studies we conducted to test both Moovy's design and implementation (UX evaluation) and

the effectiveness of the training paradigm (Moovy and Embodied Argument Movement paradigm evaluation).

4.1 UX evaluation

Firstly, we performed a UX Evaluation with the purpose to investigate: *Moovy's usability*, including the physical features such as the layout and size of the game board, the number of stimuli presented, the duration of the activities, etc; *Moovy's engagement*, as for example the ability to keep the child engaged on a prolonged period of time; *Moovy's likability*. We designed a Within-Subjects' study with two experimental conditions: condition A - session with Moovy and condition B - control with a traditional setting.

4.1.1 Target Group. The study involved 7 neurotypically Italian monolingual children, 2 girls and 5 boys, age 3.5-8 y.o. (mean 5.71, SD 1.83). Recruited children were randomly assigned to a specific order of exposure: 15 minutes of interaction with Moovy followed by 15 minutes of interaction with traditional setting (control) or vice versa.

4.1.2 Procedure. Each child performed a 30 minutes-session. Initially, the experimenter presented the TUI and the stimuli by asking the child to name and identify all of them to exclude any bias related to children's language skills and vocabulary. Afterward, she provided a brief explanation and some examples of passive sentences and sentences containing a clitic pronoun. Then each subject underwent one activity with Moovy and one with the traditional setting or vice-versa. We defined "traditional setting" as Moovy's game tabletop panel, cards, and user interaction without any digital components and feedback and without the use of the TUI. Each activity consisted of 4 (warm-up) + 12 (trial) pairs of sentences equally matched for complexity and morphosyntactic targeted structures. After each activity, the children filled in the BIS (Behaviour Intention Scale), and the UES-S (User Engagement Scale - Short version) questionnaire [32]. When necessary, the experimenter helped the children to understand the questions. All subjects were evaluated at two different times: after trying Moovy (T1) and after trying the traditional setting (T2).

4.1.3 Results and Discussion. Table 1 reports the BIS scale and UES-S questionnaire items and results. The BIS consists of a single measure of appreciation in the form of the following question: "*I would like to play this game again next weekend*" to which the subject responds by giving a score from 1 to 7 on a Likert scale. The UES-S is a 10 questions-questionnaire with a 7 points Likert scale responses. For this study, the questions were translated and adapted to the Italian language.

Since the pool of children included in the study was very small, we performed purely descriptive analyses. For each question we calculated *Mean*, *Standard Deviation* and *Median* values and compared the two experimental conditions. Results showed a greater degree of likability and engagement when the children were engaged in the activity with Moovy rather than only with cards and the game tabletop panel (traditional setting). UES-s scores were higher in questions Q1, Q4, Q6, Q7, Q9, lower in Q8, Q10 and equal in Q2, Q3 and Q5. Mean BIS score for Moovy was 7 and for the control condition 6. Moreover, all children were able to use Moovy without difficulty: they understood the meaning of all stimuli, they had no

Question	Test	Scale				Moovy	Control		
BIS- Behaviour Intention Scale									
			Mean	SD	Median		Mean	SD	Median
I would like to play this game again next weekend		1-7	4.75	2.64	6		6	1.91	7
UES-S User Engagement Scale-Short version									
Q 1	The time I spent using Application X just slipped away.	1-7	4.43	2,30	5		5	2,38	6
Q 2	My experience was rewarding.	1-7	5.29	2.06	6		6.14	0.9	6
Q 3	I felt frustrated while using this Application X	7-1	5.57	2.3	7		5.86	2.19	7
Q 4	I felt interested in this experience	7-1	4.43	2.57	5		6	1.15	6
Q 5	This Application X was attractive	1-7	4.86	1.95	5		5.29	2.14	5
Q 6	This Application X appealed to my senses	1-7	4.57	2.37	5		5.43	2.7	6
Q 7	I found this Application X confusing to use	1-7	4.57	2.51	5		5.43	2.44	7
Q 8	Using this Application X was taxing	1-7	5.43	1.72	6		3.71	2.43	3
Q 9	My experience was rewarding	1-7	4	2.31	3		5.14	2.54	6
Q 10	Using Application X was not worthwhile	1-7	5.43	1.9	6		4.71	2.21	5

Table 1: Questions scores for BIS and UES-S and results mean, SD and Median values for both Control and Moovy conditions

physical limitations in moving objects and placing them on the board, and they did not show signs of frustration or request to stop the session. However, the answers to question Q8 indicate that children found the activity with Moovy more demanding than the one with simple cards. This greater effort could indicate a greater work on the brain areas involved in both movement and language providing a small evidence that the paradigm on which Moovy was built is effective.

4.2 Moovy and Embodied Argument Movement Paradigm evaluation

The second study aimed at evaluating Moovy's effectiveness and the Embodied Argument Movement Paradigm. First, we investigated the efficacy of the TUI in facilitating the learning of the targeted morphosyntactic constructs in children with DLD. Our hypothesis was that children exposed to a training with Moovy would eventually produce more clitics and passives. Secondly, we wanted to gather insight about the soundness of the Embodied Argument Movement Paradigm we postulated. To test these aspects, we conducted a study where we compared the results of two specific tests on clitics and passives competence before and after children undergo a 4-weeks training with Moovy.

4.2.1 Target Group. The study involved 6 atypically Italian monolingual children, 2 girls and 4 boys, age 5-8 y.o. (mean 5.31, SD 3.58) with DLD. All recruited children were previously assessed with: Pre-assessment test of clitic pronouns production [3]; Pre-assessment test of passive sentence productions [42]; Morphosyntactic Comprehension Skills Assessment Test (BVL- Battery for the Assessment of Language in Children from 4 to 12 Years, [28]); Assessment test of nonverbal cognitive skills (Raven's Matrices Test,[34]).

4.2.2 Procedure. Each child was exposed to a 4-weeks training (a total of 4 training sessions), one session per week, and 2 evaluation sessions (pre and post). Each training session lasted approximately 10-15 minutes. The training activity consisted of 12 pairs of morphosyntactic targeted sentences and 4 fillers sentences with a canonical *subject-verb-object* structure (16 pairs in total). The filler

Biographical data			Clitics			Passive		
Subjects	Age	Gender	Pre	Post	Delta	Pre	Post	Delta
1	5.3	F	9/10	10/10	+ 1	10/16	10/16	0
2	6.8	M	0/10	2/10	+ 2	7/16	8/16	+ 1
3	7.7	M	6/10	9/10	+ 3	13/16	14/16	+ 1
4	8	M	6/10	9/10	+ 3	11/16	8/16	- 3
5	6.1	F	1/10	2/10	+ 1	11/16	12/16	+ 1
6	6.1	M	1/10	2/10	+ 1	11/16	14/16	+ 3

Table 2: Report of participants' biographical data and pre, post and delta scores for clitics and passives assessment.

sentences had a twofold purpose: reducing any possible bias that may be associated with the mechanical nature of the gesture that the child performs; ensuring maximum attention to stimuli. For this study we trained *only* clitic pronouns and not passive sentences so all training activities were only on clitics. The pre-assessment were performed one week before the beginning of the training and the post-assessment one week after its end.

4.2.3 Results and Discussion. The results of this pilot study are preliminary and cannot be considered conclusive. The study presents several limitations, such as the absence of a control group and the small number of participants due to the COVID-19 outbreak. Regardless, the data collected appear to be encouraging. All subjects showed improved scores in the post-assessment both for clitics and passives as reported in Table 2. Only one child had worsened performance in passives after the training. However, the delta scores are very minimal, with one or two points improvements. Moovy seems to be a promising tool to facilitate learning in children with DLD, nevertheless, its effects need to be better investigated with a wider sample group and more rigorous studies. We are aware that, for DLD children, improvements of such a hostile part of the morphology over such a short period are infrequent and hard to achieve [24]. Speech therapy programs are usually very intensive and last several months [10], so our results, although minor and

preliminary, are indeed promising. Moreover, this data offers valuable insight into the Embodied Argument Movement Paradigm we postulated. Our findings seem to confirm the soundness of the theoretical framework used to build the paradigm and consequently the TUI. The ability of Moovy to generate learning (even if minimal) in a short period of time could be evidence that supports the theoretical soundness of the paradigm. Moovy's efficacy can be explained by the increased stimulation of the common brain area, and associating a movement to the syntactic processing seems to have allowed a boost in performance that is in line with the results of similar studies in literature [39]. Moreover, subjects' performance improvements in passive sentences, even if not trained during the study, support a learning transfer effect not only from one ability to the other as already described by Thibaut et al. [39], but also from one morphosyntactic construct to another of the same nature. This phenomenon could be a neural mechanism of priming process and pre-activation of common neural resources or as short-term plasticity of the shared neural network.

5 CLOSING REMARKS

In this paper, we have proposed a new paradigm called "Embodied Argument Movement" and Moovy, a tangible user interface based on it. Moovy is a table-top game for speech therapy designed to coach children with short and intense training. It enables multi-sensory stimulation of a specific brain region responsible for the processing of motor and of a linguistic hierarchical information [19]. With our work, we would like to offer two main contributions. The first is technological as a gamified multi-sensory TUI, such as Moovy, can have multiple advantages and improve children's comprehension of critical morphosyntactic constructs. By bringing a more interactive and interesting approach to the linguistic procedures, Moovy could be a helpful instrument for speech therapy. The second is theoretical and provides meaningful contributions to the longstanding theories on how the brain has evolved and functions. The theory of the origin of human language suggests that language and action have a similar hierarchical structure and proposes that some human language constructs evolved from an ancestral motor system. The findings of our studies support such theory and confirm a common cross-domains neural structure for syntax processing. The paradigm presented in this paper may be a framework for creating new and more effective intervention tools in the field of Language Disorders.

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